

Unit 3C: Uncertainty

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1. Significant Digits

What are significant digits?

Significant digits (also called **significant figures**) are the digits in a number that give us useful information about the *accuracy* of a measurement. They tell us how carefully or exactly something has been measured.

Numbers in math are often exact (like $2 + 2 = 4$). But in real life, when we measure things like the length of a desk or the weight of rice we can never measure perfectly. That's where significant digits help us. They show which part of a number is reliable.

Rules for finding significant digits:

1. **All non-zero digits count.**
 - Example: 123 has 3 significant digits.
 - Example: 8.976 has 4 significant digits.
2. **Zeros between non-zero digits count.**
 - Example: 3005 has 4 significant digits.
 - Example: 20.08 has 4 significant digits.
3. **Leading zeros (zeros before non-zero numbers) never count.**
 - Example: 0.0049 has 2 significant digits (4 and 9).
 - Example: 0.00056 has 2 significant digits (5 and 6).
4. **Trailing zeros (zeros at the end) count only if there is a decimal.**
 - Example: 2500 has 2 significant digits.
 - Example: 2500. has 4 significant digits (the decimal makes the zeros meaningful).
 - Example: 25.00 has 4 significant digits.

Example:

- A classroom ruler can measure up to 0.1 cm. If you measure your pencil as **14.2 cm**, that has 3 significant digits.
- A carpenter's measuring tape may only measure up to 1 cm. The same pencil may be recorded as **14 cm** (2 significant digits).
- A digital lab scale can measure 125.346 g (6 significant digits), but a kitchen scale might just show 125 g (3 significant digits).

So, the more significant digits a number has, the more precise the measurement is.

2. Random and Systematic Errors

What is an error?

When we measure something, we almost never get the *true value*. The difference between what we measure and the true value is called an **error**. Errors are not the same as mistakes. A mistake is when you do something wrong, but an error happens naturally because measuring is never perfect.

Random Errors

- These errors happen by **chance** and are unpredictable.
- They can make the measurement a little too high or a little too low.
- If you repeat the measurement many times, the results scatter around the true value.

Examples:

1. Timing a race with a stopwatch. Sometimes you press stop too early, sometimes too late.
2. Measuring water with a beaker. You might look at it from a slightly different angle each time, making the reading a bit higher or lower.
3. Listening to a clock tick. Sometimes you might mishear and count one extra tick.

Systematic Errors

- These errors happen in the **same direction every time**.
- They usually come from a problem in the tool or the method used.
- They shift all results too high or too low.

Examples:

1. A bathroom scale that is not reset to zero shows 2 kg more every time.
2. A thermometer that is wrongly calibrated and always shows 5°C less than the real temperature.
3. Using a stretched measuring tape that is longer than it should be, so everything you measure comes out shorter than the true value.

Why this matters:

- Random errors can be reduced by repeating the experiment and averaging.
- Systematic errors cannot be fixed by repeating. You must change or fix the tool.

3. Errors in Absolute and Relative Terms

Absolute Error

This is the difference between the measured value and the actual (true) value.

Formula:

$$\text{Absolute Error} = |\text{Measured Value} - \text{True Value}|$$

Example 1:

- True length of a rod = 50.0 cm
- Measured length = 49.2 cm
- Absolute error = $|49.2 - 50.0| = 0.8$ cm

Example 2:

- True weight of flour bag = 2.0 kg
- Measured = 2.1 kg
- Absolute error = $|2.1 - 2.0| = 0.1$ kg

Relative Error

This shows how big the error is compared to the true value. It is usually written as a fraction or percentage.

Formula:

$$\text{Relative Error} = \text{Absolute Error} \div \text{True Value}$$

Example 1:

- Absolute error = 0.8 cm
- True value = 50.0 cm
- Relative error = $0.8 \div 50 = 0.016 = 1.6\%$

Example 2:

- Absolute error = 0.1 kg
- True value = 2.0 kg
- Relative error = $0.1 \div 2 = 0.05 = 5\%$

Why use both?

Absolute error shows the actual difference.

Relative error shows how important that difference is.

- If you miss 1 cm when measuring a 2 cm button, that's a huge mistake (50%).
- If you miss 1 cm when measuring a 10 m rope, that's tiny (0.01%).

So relative error helps us judge whether an error really matters.

4. Accuracy and Precision

Accuracy

How close a measurement is to the true value.

Precision

How close repeated measurements are to each other, even if they are not close to the true value.

Example:

1. Throwing darts at a dartboard:

- If darts hit near the bull's-eye, you are accurate.
- If darts land close to each other, you are precise.

2. Weighing sugar for a recipe:

- The true weight should be 100 g.
- If your scale shows 100 g, 101 g, 99 g → Accurate and fairly precise.
- If it shows 110 g, 111 g, 110 g → Precise but not accurate.
- If it shows 100 g, 90 g, 110 g → Neither accurate nor precise.

3. Shooting arrows:

- A trained archer hits the bull's-eye every time → Accurate and precise.
- A beginner hits the same wrong spot → Precise but not accurate.
- A child sprays arrows everywhere → Neither accurate nor precise.

In science and daily life:

We want both accuracy and precision.

- A doctor's thermometer must be accurate (to show the true temperature) and precise (to give consistent readings each time).
- A clock must be accurate (to show the correct time) and precise (to tick regularly).

5. Rounding Rules for Combining Approximate Numbers

When we do calculations with measurements, we must round so that the result does not look more exact than the least exact measurement.

Rule for Addition and Subtraction:

- Look at the number with the least number of decimal places.
- Round the final answer to that same number of decimal places.

Examples:

1. $12.45 \text{ cm} + 3.2 \text{ cm} = 15.65 \text{ cm}$
 - 3.2 has 1 decimal place.
 - Final Answer = 15.7 cm
2. $56.78 \text{ g} - 4.3 \text{ g} = 52.48 \text{ g}$
 - 4.3 has 1 decimal place.
 - Final Answer = 52.5 g

Rule for Multiplication and Division:

- The result should have the same number of significant digits as the number with the least significant digits.

Examples:

1. $4.56 \times 1.4 = 6.384$
 - 4.56 has 3 significant digits.
 - 1.4 has 2 significant digits.
 - Final Answer = 6.4 (2 significant digits).
2. $25.0 \div 4.52 = 5.530973\dots$
 - 25.0 has 3 significant digits.
 - 4.52 has 3 significant digits.
 - Final Answer = 5.53 (3 significant digits).

Example:

Imagine you are baking a cake:

- You weigh flour as 1.2 kg (2 significant digits).
- Sugar is 0.845 kg (3 significant digits).

- When adding, your answer should have only 2 significant digits because flour was less exact.
So total = 2.0 kg (not 2.045 kg).

This prevents you from giving a result that looks more exact than your least careful measurement.